

# Numerical validation of `finite_contour_cocycles_short.tex`

Companion to `validate.py`

## 1 What this suite is for

Every closed form in the note is a claim that some explicitly-summable expression equals a contour integral. Algebra alone does not catch a claim that is right in one geometric regime and wrong in another: the error is invisible until the wrong regime is evaluated. This suite therefore takes *raw quadrature of the defining contour integral* as ground truth and checks each closed form against it, never against another closed form.

Two design rules follow.

**Index by regime, not by equation.** Each check runs across the geometric configurations the note distinguishes: a pole above the  $T$ -segment, on the edge  $\text{Im } z = 1$  with  $z \neq i$ , at  $i$ , below the contour, and absent. A per-equation test that samples one configuration certifies nothing about the others.

**Ground truth must be exact.** Where a check needs Fourier coefficients, they come from integer  $q$ -series arithmetic, not from numerical extraction. Recovering  $c_f(n)$  from  $\int_0^1 f(x+iy)e^{-2\pi in(x+iy)}dx$  is catastrophically ill-conditioned for a form with a pole at the cusp: for  $1/\Delta$  the integrand runs as  $e^{2\pi ny}$  while the answer is only  $e^{4\pi\sqrt{n}}$ , so  $c(18)$  at  $y = 1.3$  asks for a  $10^{23}$  number out of  $10^{67}$ -sized values. The suite computes  $\prod(1 - q^n)^{\pm 24}$  in exact integer arithmetic instead; `cf_Delta` and `cf_invDelta` reproduce  $\tau(n)$  and the known  $1/\Delta = q^{-1} + 24 + 324q + 3200q^2 + \dots$  exactly.

Each check carries the `\label` it validates, so a failure names a line in the note rather than a formula in Python. Two tiers: `validate.py` (coarse precision, one case per regime) and `validate.py --slow` (high precision, full cross-product). A third tier, `report`, prints a measured discrepancy without passing or failing; it is used for questions the note has not yet settled, so the suite documents them instead of presuming an answer.

## Shared machinery in validate.py

|                                      |                                                                                                 |
|--------------------------------------|-------------------------------------------------------------------------------------------------|
| <code>ktil(tau,s,k)</code>           | the Hurwitz $T$ -kernel $\tilde{k}_T$ of <code>def:kernel</code>                                |
| <code>LT_raw(f,k,s,tau0)</code>      | <b>ground truth:</b> quadrature of $e^{-i\pi s/2} \int_{\tau_0-1}^{\tau_0} f \tilde{k}_T d\tau$ |
| <code>LT_closed(cf,k,s,tau0)</code>  | the closed form <code>eq:T-segment-closed-Mk</code>                                             |
| <code>E4, E6, Delta, invDelta</code> | the forms as functions on $\mathbb{H}$ , for quadrature                                         |
| <code>cf_E4, cf_E6, cf_Delta</code>  | their <i>exact</i> integer $q$ -coefficients                                                    |
| <code>cf_invDelta, cf_j, jay</code>  | likewise for $1/\Delta$ and $j = E_4^3/\Delta$                                                  |
| <code>residue_at(g,s0,r)</code>      | Res by trapezoid rule on a circle $ s - s_0  = r$                                               |
| <code>WEAK_FORMS</code>              | the Layer C form table (form, function, coefficients, $k$ )                                     |
| <code>_eta_prod, _smul, _sinv</code> | integer power-series arithmetic behind those                                                    |
| <code>check(label,layer,tier)</code> | decorator registering a test against a <code>\label</code>                                      |
| <code>rel(got,want)</code>           | relative error, denominator floored at 1                                                        |
| <code>main</code>                    | driver; TOL, NT, NMAX are set by tier                                                           |

## 2 Layer A: kernel and unfolding

Layer A sits upstream of everything else. If the kernel identities or the mode-by-mode unfolding are wrong, every downstream check is measuring the wrong thing.

**A1 — the  $\delta_T$  shift** (`def:kernel; a1_kernel_shift`). Verifies  $\delta_T \tilde{k}_T(\tau, s) = \tilde{k}_T(\tau + 1, s) - \tilde{k}_T(\tau, s) = -(\tau + 1)^{s-1} + e^{i\pi(s-1)}(\tau + 1)^{k-1-s}$  directly, at three weights, three base-points and three values of  $s$  including one on the critical line. This is the identity that turns the  $\partial_{\tau_0}$  boundary term into a unit shift of the kernel, so it underwrites  $\tau_0$ -independence.

**A2 — the Hurwitz antiderivative** (`eq:hurwitz-antideriv; a2_hurwitz`). Three separate claims:  $\partial_a [\zeta(\sigma - 1, a)/(1 - \sigma)] = \zeta(\sigma, a)$ ; the shift  $\zeta(\sigma - 1, a + 1) - \zeta(\sigma - 1, a) = -a^{1-\sigma}$ ; and the definite integral  $\int_{i-1}^i \zeta(\sigma, \tau + 1) d\tau = i^{1-\sigma}/(\sigma - 1)$  that they combine to give. These produce the  $n = 0$  contribution, hence the entire  $-c_f(0)(\frac{1}{s} + \frac{i^k}{k-s})$  polar structure.

**A3 — the unfolding** (`eq:T-segment-closed-Mk`). The general- $\tau_0$   $T$ -segment closed form, checked against quadrature in three stages of increasing exposure:

- `a3a_unfold_pos` — positive modes only ( $\Delta, k = 12$ ): no branch question arises;
- `a3b_unfold_const` —  $c_f(0) \neq 0$  ( $E_4, k = 4$ ): exercises the  $-c_f(0)((\tau_0/i)^s/s + \dots)$  term;
- `a3c_unfold_neg` — negative modes ( $1/\Delta, k = -12$ ):  $n < 0$  makes  $(2\pi n)^s$  a power of a negative number and drives  $-2\pi i n \tau_0$  onto the cut of  $\Gamma(s, \cdot)$ .

The third stage is the “incomplete-gamma unfolding ambiguity”. The suite pins the convention by quadrature: for  $n < 0$ ,

$$(2\pi n)^s := (2\pi |n|)^s e^{+i\pi s}.$$

This must be applied as an explicit phase and never folded into the base as  $(2\pi |n| e^{i\pi})^s$ : numerically  $e^{i\pi} = -1 + \varepsilon i$  with the sign of  $\varepsilon$  a rounding artifact, so the base-folded form silently changes branch with the working precision. With the phase explicit, the closed form matches quadrature to  $10^{-31}$  at  $\tau_0 = i, 1.35i, 2i$  and at  $\tau_0$  off-axis to the right.

**A3,  $\text{Re } \tau_0 < 0$ : the continued  $\Gamma$  (a3d\_unfold\_neg\_left).** For  $n < 0$  write  $z = -2\pi i n \tau_0$ . With  $\tau_0 = x + iy$  this is  $z = -2\pi |n|y + 2\pi i |n|x$ , so  $\text{Re } z < 0$  always while  $\text{Im } z$  carries the sign of  $x = \text{Re } \tau_0$ : the point  $z$  sits just above the negative-real cut of  $\Gamma(a, \cdot)$  when  $\text{Re } \tau_0 > 0$  and just below it when  $\text{Re } \tau_0 < 0$ . The L-integral varies continuously as  $\tau_0$  moves; the *principal*  $\Gamma$  jumps.

The jump is the standard monodromy. Since  $\Gamma(a, z) = \Gamma(a) - \gamma(a, z)$  with  $\gamma(a, z) = z^a \sum_{m \geq 0} (-z)^m / (m!(a+m))$ , all the multivaluedness sits in  $z^a$ , and one anticlockwise turn gives

$$\Gamma_{\text{cont}}(a, z) = \Gamma(a)(1 - e^{2\pi i a}) + e^{2\pi i a} \Gamma_{\text{prin}}(a, z),$$

implemented as `_gamma_cont` and applied per term at  $a = s$  and  $a = k - s$ . With it, `eq:T-segment-closed-Mk` reproduces quadrature to  $10^{-31}$  for  $\text{Re } \tau_0 < 0$ , just as the principal branch does for  $\text{Re } \tau_0 \geq 0$ .

So the lemma is correct as stated for all  $\tau_0 \in \mathbb{H}$ , and `prop:HGG`'s phrase “the identity continues to  $n < 0$ ” is doing real work:  $\Gamma$  must be the continuation, which coincides with the principal value only while  $\text{Re } \tau_0 \geq 0$ . An earlier draft of this document reported the lemma as failing for  $\text{Re } \tau_0 < 0$ ; that was an artifact of evaluating  $\Gamma$  on the principal branch on the far side of its cut, not a defect of the result.

### 3 Layer B: $\tau_0$ -independence and wall-crossing

Layer B tests the full L-integral  $L^* = L_S^* + L_T^*$  of `def:Lint`, with both segments by quadrature (`LS_raw`, `LT_raw_path`, assembled by `L_raw`). Paths are polylines integrated by `path_int`;  $\mathbb{H}$  is convex, so the straight line from  $-1/\tau_0$  to  $\tau_0$  stays inside it.

**B1, B2 —  $\tau_0$ -independence (b1\_indep\_onaxis, b2\_indep\_offaxis).** `prop:indep`: neither segment is separately invariant, only the sum. Checked on the imaginary axis and off it, for  $\Delta$ ,  $E_4$ ,  $E_6$  and  $1/\Delta$ . B2 includes  $\text{Re } \tau_0 < 0$  deliberately, which is what first showed that the A3 discrepancy could not be a property of the L-integral: the raw  $L^*$  is invariant there to  $10^{-16}$ , so whatever was wrong had to lie in the evaluation of the closed form — as it did.

**B3 — a meromorphic form (b3\_indep\_meromorphic).** `prop:indepHomotopy` for  $f = \Delta/(j - j(2i)) \in F_{12}$ , whose poles are the  $\Gamma$ -orbit of  $2i$  at heights  $2, \frac{1}{2}, \frac{2}{5}, \dots$ . Base-points near  $i$  keep both segments clear of every image, so these all lie in one class and  $L^*$  must not move.

**B4, B5 — wall-crossing (b4\_wall\_T, b5\_wall\_S).** `lem:wall` with *fixed end-points and two homotopy classes*, so the crossing is isolated from the genuine  $\tau_0$ -dependence of each separate segment. For  $T$ : a straight segment at height 2.2 against one dipping to 1.8, enclosing  $2i$  alone. For  $S$ : two paths agreeing around every image except  $2i$ , which one passes to the left and the other to the right. In both the enclosed loop runs clockwise, so  $X = -1$  and the predicted jump is  $-2\pi i e^{-i\pi s/2} r$ , with  $r_T = \text{Res}(f \tilde{k}_T)$  and  $r_S = \text{Res}(f \tau^{s-1})$  from `res_circle`.

Two things had to be got right for these to test anything. The form is normalised so  $\text{Res}_{2i} f = 1$ ; unnormalised the residue is  $\Delta(2i)/j'(2i) \approx 2 \times 10^{-12}$  and every jump is  $\sim 10^{-9}$ , at which scale a sign error hides under the noise floor. And `res_circle` needs enough points: the trapezoid rule on  $|\tau - 2i| = r$  converges like  $(r/R)^M$  with  $R = 1$  the distance to  $2i + 1$ , so the original  $r = 0.3$ ,  $M = 16$  gave only  $0.3^{16} \sim 4 \times 10^{-9}$  — agreement to eight digits, which reads as a failure against a  $10^{-9}$  tolerance and invites a hunt for a nonexistent bug elsewhere.

## 4 Layer C: `thm:weakL` and the functional equation

For  $f \in M_k^!$  the base-point may be taken at  $\tau_0 = i$ , where the  $S$ -segment degenerates to a point and  $L^* = L_T^*$  alone. Ground truth for this layer is therefore `LT_raw(f,k,s,I)`, the bare quadrature of  $e^{-i\pi s/2} \int_{i-1}^i f \tilde{k}_T d\tau$ .

The five test forms (`WEAK_FORMS`) are chosen to span the cases the statement distinguishes rather than to be interesting individually:

| form       | code                     | $k$ | feature exercised                           |
|------------|--------------------------|-----|---------------------------------------------|
| $\Delta$   | <code>cf_Delta</code>    | 12  | cuspidal form, $c_f(0) = 0$                 |
| $E_4$      | <code>cf_E4</code>       | 4   | $c_f(0) \neq 0$ , so the polar term is live |
| $E_6$      | <code>cf_E6</code>       | 6   | $i^k = -1$                                  |
| $1/\Delta$ | <code>cf_invDelta</code> | -12 | pole at the cusp, so $n < 0$ modes          |
| $j$        | <code>cf_j</code>        | 0   | the $s = 0$ and $s = k$ poles coincide      |

$j = E_4^3/\Delta$  is built by integer series arithmetic and reproduces  $q^{-1} + 744 + 196884q + 21493760q^2$  exactly;  $E_6$  reproduces  $1 - 504q - 16632q^2 - 122976q^3$ .

**C1 — `eq:weakL` (`c1_weakL`).** The closed form against quadrature, for all five forms. Since `eq:weakL` is `eq:T-segment-closed-Mk` specialised to  $\tau_0 = i$ , this also re-exercises the branch convention pinned in A3, now with  $c_f(0) \neq 0$  and with  $k \leq 0$ .

**C2 — the residue (`c2_residue`).**  $\text{Res}_{s=0} L^* = -c_f(0)$ . The residue is extracted from the *raw* integral by a contour in  $s$  (`residue_at`, trapezoid rule on  $|s| = r$ ), so the note's polar claim is tested against quadrature and not against its own closed form. The circle is used rather than extrapolating  $s L^*(s)$ , which converges only linearly in  $s$  and would not reach tolerance.

The weight-zero case is the sharp one. For  $k = 0$  the two poles collide, and the polar term is  $-c_f(0)(\frac{1}{s} + \frac{i^0}{0-s}) = 0$  identically, so the residue must *vanish* even though  $c_j(0) = 744$ . It does. This is the mechanism behind the factor  $(1 - \delta_{k,0})$  appearing in the special-value corollaries of arXiv:1806.09874, recovered here from bare quadrature.

**C3 — the functional equation (`c3_functional_eq`).**  $L^*(f, s) = i^k L^*(f, k - s)$ , with *both* sides computed by raw quadrature. This is the cheapest global probe in the suite: it touches every phase in the construction at once, and a single lost sign anywhere breaks it.  $E_6$  is the case that matters, being the only form here with  $i^k = -1$ ; the weights 12, 4, -12, 0 all give  $i^k = +1$  and would not detect a dropped  $i^k$ .

## 5 Layers B, D–G

Pending, in dependency order: **B**  $\tau_0$ -independence and wall-crossing; **D** periods, quasi-periods and the Haberland–Petersson pairing; **E** the meromorphic machinery of §4; **F** external anchors (the corrected 1806 lemma, the Hurwitz class-number sum rule); **G** suite hygiene — truncation, precision and quadrature self-tests.